LEARNER GUIDE

Inequalities as regions

Equations, Inequalities and Simultaneous Relationships · Lesson 11. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

On a number line an inequality was a range. In two variables it is a whole **region** of the plane. The boundary is a line: **dashed** if the inequality is strict (< or >, the line is not included) and **solid** if it is inclusive (≤ or ≥). The line splits the plane in two; the solutions are all the points on **one** side. To find which side, **test a point** — the origin (0, 0) is easiest — and shade the side that makes the inequality true. When two inequalities must both hold, the answer is where the two regions **overlap**.

What you are learning to notice

- A two-variable inequality is a **region** (a half-plane), not a line.
- Boundary **dashed** for strict (< >), **solid** for inclusive (≤ ≥).
- One **test point** (the origin, if it is not on the line) decides which side to shade.
- Two inequalities together → the **overlap** of their regions.

Moving through it, one step at a time

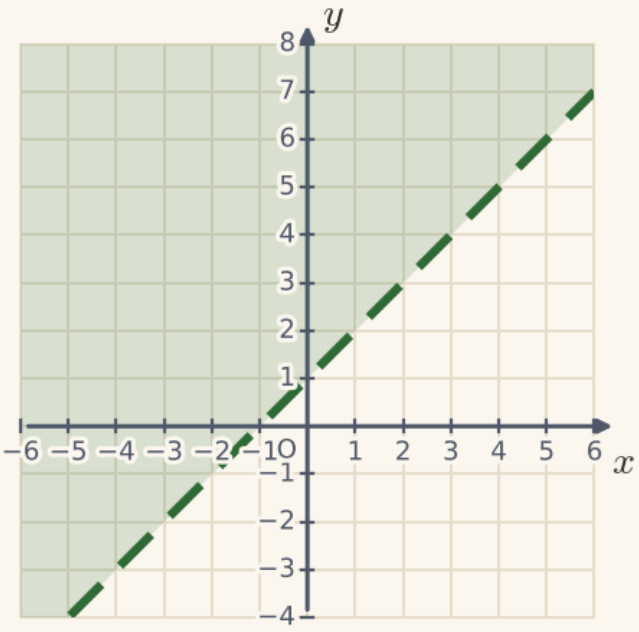
Draw the boundary line (from y = mx + c, or a vertical line x = a). Make it dashed for a strict inequality and solid for an inclusive one. Choose a test point not on the line — usually (0, 0) — and substitute it: if the inequality is true there, shade that side; if not, shade the other side. For two inequalities, shade each and take the region they share. Check by testing a point inside (should work) and one outside (should not).

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Shade a region

Shade the region $y > x + 1$.

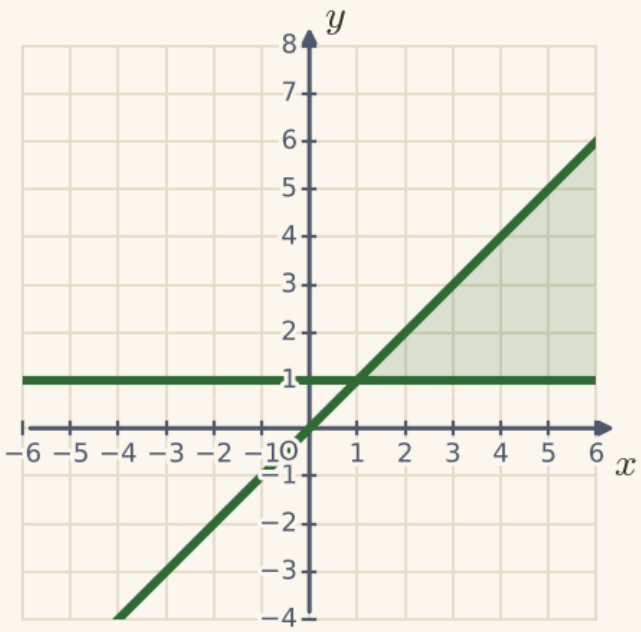


1. Draw the boundary $y = x + 1$, dashed (> is strict).	dashed line
2. Test the origin: is $0 > 0 + 1$?	$0 > 1$ is false
3. So shade the OTHER side — above the line.	region shaded above

$y > x + 1$ is the half-plane above the dashed line $y = x + 1$ (the origin is below it, and $0 > 1$ is false).

Example 2 — Where two regions overlap

Shade the region where $y \geq 1$ AND $y \leq x$.



1. $y \geq 1$: solid horizontal boundary, shade above.	above $y = 1$
2. $y \leq x$: solid boundary $y = x$, shade below.	below $y = x$
3. The answer is where both hold — the overlap.	the wedge between them

The overlap is the wedge where $1 \leq y \leq x$ — exactly the compound-region idea from Constructions, built from straight boundaries.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (plotting $y = mx + c$, solving inequalities and open/closed ends (Lesson 10), and the idea of a region from the Constructions unit), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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